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The law of the iterated logarithm for discrepancies of $\{\theta^n x\}$

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Dedicated to the memory of Professor Walter Philipp

Abstract. It is proved that two types of discrepancies of the sequence $\{\theta^n x\}$ obey the law of the iterated logarithm with the same constant. The appearing constants are calculated explicitly for most of $\theta > 1$.

1. Introduction

We recall the definition of the two types of discrepancies of a sequence $\{x_k\}$ of real numbers:

$$D_N(\{x_k\}) := \sup_{0 \leq a' < a < 1} \left| \frac{1}{N} \sum_{k=1}^N \mathbf{1}_{[a', a)}(\langle x_k \rangle) - (a - a') \right|;$$
$$D_N^*(\{x_k\}) := \sup_{0 \leq a < 1} \left| \frac{1}{N} \sum_{k=1}^N \mathbf{1}_{[0, a)}(\langle x_k \rangle) - a \right|,$$

where $\mathbf{1}_{[a', a)}$ denotes the indicator function of $[a', a)$ and $\langle x \rangle$ denotes the fractional part $x - [x]$ of real number x . The inequality $D_N^* \leq D_N \leq 2D_n^*$ is trivial.

The sequence $\{x_k\}$ is uniformly distributed mod 1, if and only if $D_N \rightarrow 0$ as $N \rightarrow \infty$. When the sequence $\{n_k\}$ satisfies the Hadamard's gap condition $n_{k+1}/n_k > q > 1$, it is known that $\{n_k x\}$ is uniformly distributed mod 1 for almost every x . Philipp [5, 6] proved the following asymptotic property and solved the Erdős-Gál conjecture.

$$\frac{1}{4\sqrt{2}} < \overline{\lim}_{N \rightarrow \infty} \frac{ND_N(\{n_k x\})}{\sqrt{2N \log \log N}} \leq 166 + 664/(q^{1/2} - 1) \quad \text{a.e.}$$

For special sequence $\{2^k\}$, Philipp [4] proved the law of the iterated logarithm:

$$\overline{\lim}_{N \rightarrow \infty} \frac{ND_N(\{2^k x\})}{\sqrt{2N \log \log N}} = \sup_{0 \leq a' < a < 1} \sigma_{2, a', a} \in [\sqrt{42}/9, \sqrt{2}] \quad \text{a.e.,}^*$$

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* In [4], it is written $\sqrt{48}/9$ instead of $\sqrt{42}/9$. It is apparently a typographical error, since our Corollary proves that the supremum equals to $\sqrt{42}/9$

where $\sigma_{2,a',a}$ is given as a part of definition of $\sigma_{\theta,a',a}$ ($\theta > 1$) below: Put $f_{a',a}(x) = \mathbf{1}_{[a',a)}(\langle x \rangle) - (a - a')$. If $\theta^r \notin \mathbf{Q}$ for all $r \in \mathbf{N}$, then

$$\sigma_{\theta,a',a}^2 := \int_0^1 f_{a',a}^2(x) dx.$$

If $\theta = \sqrt[r]{p/q}$ with $p, q \in \mathbf{N}$, $r = \min\{k \in \mathbf{N} \mid \theta^k \in \mathbf{Q}\}$, and $\gcd(p, q) = 1$,

$$\sigma_{\theta,a',a}^2 := \int_0^1 f_{a',a}^2(x) dx + 2 \sum_{k=1}^{\infty} \int_0^1 f_{a',a}(p^k x) f_{a',a}(q^k x) dx.$$

Our purpose is to extend the result of Philipp and determine the asymptotic behaviour of the discrepancies of $\{\theta^n x\}$ for every $\theta > 1$.

Theorem. For $\theta > 1$,

$$\Sigma_{\theta} := \overline{\lim}_{N \rightarrow \infty} \frac{ND_N(\{\theta^k x\})}{\sqrt{2N \log \log N}} = \overline{\lim}_{N \rightarrow \infty} \frac{ND_N^*(\{\theta^k x\})}{\sqrt{2N \log \log N}} = \sup_{0 \leq a < 1} \sigma_{\theta,0,a}, \quad \text{a.e. } x.$$

Here we emphasise that two discrepancies D_N^* and D_N are proved to have the same asymptotic behaviour. By calculating $\sup_{0 \leq a < 1} \sigma_{\theta,0,a}$ explicitly, we have the results below:

Corollary.

- (1) When $\theta^r \notin \mathbf{Q}$ for all $r \in \mathbf{N}$, $\Sigma_{\theta} = 1/2$, a.e.
- (2) When $\theta = \sqrt[r]{p/q}$ with $p, q \in \mathbf{N}$, $r = \min\{k \in \mathbf{N} \mid \theta^k \in \mathbf{Q}\}$, and $\gcd(p, q) = 1$, $1/2 \leq \Sigma_{\theta} \leq \sqrt{(pq+1)/(pq-1)}/2$, a.e.
- (3) And moreover, if p and q are odd numbers, $\Sigma_{\theta} = \sqrt{(pq+1)/(pq-1)}/2$, a.e. Especially, when p is odd and $q = 1$, $\Sigma_{\theta} = \sqrt{(p+1)/(p-1)}/2$, a.e.
- (4) When $p \geq 4$ is even and $q = 1$, $\Sigma_{\theta} = \sqrt{(p+1)p(p-2)/(p-1)^3}/2$, a.e.
- (5) When $p = 2$ and $q = 1$, $\Sigma_{\theta} = \sqrt{42}/9$, a.e.
- (6) When $p = 5$ and $q = 2$, $\Sigma_{\theta} = \sqrt{22}/9$, a.e.

By (3), (4), and (5), we have completed the explicit calculation of Σ_{ν} for every integer $\nu \geq 2$. The contrast between even case and odd case seems very interesting, and the isolation of the number 2 is mysterious. Although (6) seems very incomplete result, we present here for it is proved by completely different method as compared to the previous ones.

Except for the last case, we could not calculate Σ_{ν} explicitly when $p/q \notin \mathbf{N}$ and pq is even.

2. The law of the iterated logarithm for Riesz-Raikov sums

In this section, we simply denote $f := f_{a',a}$ and $\sigma_f := \sigma_{\theta,a',a}$. Let s_n be the n -th partial sum of the Fourier series of f . We have $R(n) := \|f - s_n\|_2 = O(n^{-1/2})$. Put $a_{M,N} := \int_0^1 \left(\sum_{k=M+1}^{M+N} f(\theta^k x) \right)^2 dx$. If we prove

$$\sup_M \left| \frac{a_{M,N}}{N} - \sigma_f^2 \right| \rightarrow 0, \quad (2.1)$$

by following the argument given by Berkes [1: part II] (the proof of Remark 2 a), i.e., the case $n_{k+1}/n_k \rightarrow \infty$), we can derive the law of the iterated logarithm (Corollary 2 of [1: Part II])

$$\overline{\lim}_{N \rightarrow \infty} \frac{1}{\sqrt{2N \log \log N}} \sum_{k=1}^N f(\theta^k x) = \sigma_f, \quad \text{a.e.} \quad (2.2)$$

when $\sigma_f^2 > 0$. As we prove later, it remains valid when $\sigma_f^2 = 0$.

By the method of [2, 3], we prove (2.1). Put

$$A_{l,k}(f) := \int_0^1 f(\theta^k x) f(\theta^{k+l} x) dx.$$

By noting $\int_0^1 |g(\theta^k x)|^2 dx \leq 2\|g\|_2^2$ and by applying Schwarz inequality, we have $|A_{l,k}(f) - A_{l,k}(s_n)| \leq 4\|f - s_n\|_2 \|f\|_2$. By noting $\|f\|_2 \leq 1$ and $R(n) = O(n^{-1/2})$, and by applying Lemma 5.1 of [1: Part I], we have

$$|A_{l,k}(f)| \leq C\theta^{-l/2} \|f\|_2.$$

If $\theta^l = \nu/\mu$ ($\mu, \nu \in \mathbf{N}$), we have

$$A_{l,k}(f) = \frac{1}{\theta^k/\mu} \int_0^{\theta^k/\mu} f(\mu x) f(\nu x) dx \xrightarrow{k \rightarrow \infty} \int_0^1 f(\mu x) f(\nu x) dx.$$

If $\theta^l \notin \mathbf{Q}$, then

$$A_{l,k}(s_n) = \sum_{|m|, |m'| \leq n} \widehat{f}(m) \widehat{f}(m') \frac{\exp(2\pi i \theta^k (m + m' \theta^l)) - 1}{2\pi i \theta^k (m + m' \theta^l)} \xrightarrow{k \rightarrow \infty} 0.$$

Hence $|A_{l,k}(f)| \leq 4\|f\|_2 \|f - s_n\|_2 + |A_{l,k}(s_n)| \rightarrow 2\|f\|_2 \|f - s_n\|_2$ as $k \rightarrow \infty$, and thereby $A_{l,k}(f) \rightarrow 0$ ($k \rightarrow \infty$). Hence we have proved

$$A_{l,k}(f) \rightarrow A_{l,\infty}(f) := \begin{cases} 0 & \text{if } \theta^l \notin \mathbf{Q}, \\ \int_0^1 f(\mu x) f(\nu x) dx & \text{if } \theta^l = \nu/\mu. \end{cases}$$

In usual way, we can prove $\sup_M |\frac{1}{N} \sum_{k=M+1}^{M+N} A_{l,k}(f) - A_{l,\infty}| \rightarrow 0$ as $N \rightarrow \infty$. Put $B_l^{M,N} := \frac{1}{N} \sum_{k=M+1}^{M+N-l} A_{l,k}(f)$ for $l < N$ and $B_l^{M,N} := 0$ for $l \geq N$. By

$$|B_l^{M,N} - A_{l,\infty}| \leq \frac{N-l}{N} \left| \frac{1}{N-l} \sum_{k=M+1}^{M+N-l} A_{l,k}(f) - A_{l,\infty} \right| + \frac{l}{N} |A_{l,\infty}|,$$

we can verify

$$\lim_{N \rightarrow \infty} \sup_M |B_l^{M,N} - A_{l,\infty}| = 0 \quad \text{and} \quad |B_l^{M,N}| \leq C\theta^{-l/2} \|f\|_2.$$

Take L such that $2 \sum_{l=L+1}^{\infty} C\theta^{-l/2} \|f\|_2 < \varepsilon/2$. Because of $\frac{1}{N} a_{M,N} = B_0^{M,N} + 2 \sum_{l=1}^{N-1} B_l^{M,N}$, if $N > L$ is satisfied, $|\frac{1}{N} a_{M,N} - \sigma_f^2|$ is bounded from above by

$$|B_0^{M,N} - A_{0,\infty}(f)| + 2 \sum_{l=1}^L |B_l^{M,N} - A_{l,\infty}(f)| + 2 \sum_{l=L+1}^{\infty} (|B_l^{M,N}| + |A_{l,\infty}(f)|).$$

Since the last term is bounded by ε , we have $\overline{\lim}_{N \rightarrow \infty} \sup_M |\frac{1}{N} a_{M,N} - \sigma_f^2| \leq \varepsilon$. Because (2.1) is verified, we have (2.2) when $\sigma_f > 0$.

Finally, let us consider the case $\sigma_f = 0$. We recall the following expression [2, 3]:

$$\sigma_f^2 = \begin{cases} 2 \sum_{j \in P_p} |\sum_{s=0}^{\infty} \widehat{f}(jp^s)|^2, & \text{if } q = 1, \\ 2 \sum_{j \in P_{p,q}} \sum_{i=0}^{\infty} |\sum_{s=0}^i \widehat{f}(jp^s q^{i-s})|^2, & \text{if } q > 1, \end{cases}$$

where $P_p := \{n \in \mathbf{N} \mid p \nmid n\}$ and $P_{p,q} := P_p \cap P_q$. By this, we see that $\sigma_{f+\varepsilon \cos 2\pi x} = \varepsilon/\sqrt{2}$ if $\sigma_f = 0$ and $\varepsilon > 0$. Therefore, by applying the result (2.2), we have

$$\overline{\lim}_{N \rightarrow \infty} \frac{1}{\sqrt{2N \log \log N}} \sum_{k=1}^N (f(\theta^k x) + \varepsilon \cos(2\pi \theta^k x)) = \varepsilon/\sqrt{2}.$$

Combining this together with

$$\overline{\lim}_{N \rightarrow \infty} \frac{1}{\sqrt{2N \log \log N}} \left| \sum_{k=1}^N \varepsilon \cos(2\pi \theta^k x) \right| = \varepsilon/\sqrt{2},$$

and letting $\varepsilon \rightarrow 0$, we have (2.2) with $\sigma_f = 0$.

3. The law of the iterated logarithm for discrepancies

We follow the method of Philipp [5]. Although it is assumed that $\{n_k\}$ is the sequence of integers in [5], by using the measure $h(dx) := (\frac{\sin x}{x})^2 dx$ in the proof of Proposition of [5], we have the same conclusion without this assumption. (See Berkes [1: Part I], part c of proof of Lemma (3.4)).

After Philipp, for N , let us put $H := [\log N / \log 4] + 1$. Let us fix $L \in \mathbb{N}$ and $I = 0, \dots, 2^L - 1$ arbitrarily. For any $a \in [0, 1/2^L]$, by expanding $a = \sum_{j=L+1}^{\infty} 2^{-j} \varepsilon_j$, we have $\sum_{j=L+1}^H 2^{-j} \varepsilon_j \leq a < \sum_{j=L+1}^H 2^{-j} \varepsilon_j + 2^{-H}$. Hence, by putting $\rho_h(x) := \mathbf{1}_{[2^{-L}I + \sum_{j=L+1}^h 2^{-j} \varepsilon_j, 2^{-L}I + \sum_{j=L+1}^{h+1} 2^{-j} \varepsilon_j)}(\langle x \rangle)$ and $\sigma_H(x) := \mathbf{1}_{[2^{-L}I + \sum_{j=L+1}^H 2^{-j} \varepsilon_j, 2^{-L}I + \sum_{j=L+1}^H 2^{-j} \varepsilon_j + 2^{-H})}(\langle x \rangle)$, we have an inequality $\sum_{h=L+1}^{H-1} \rho_h(x) \leq \mathbf{1}_{[2^{-L}I, 2^{-L}I+a)}(\langle x \rangle) \leq \sum_{h=L+1}^{H-1} \rho_h(x) + \sigma_H(x)$. By denoting $\tilde{\rho}_h(x) := \rho_h(x) - 2^{-h}$ and $\tilde{\sigma}_H(x) := \sigma_H(x) - 2^{-H}$, we can derive

$$\sum_{h=L}^{H-1} \tilde{\rho}_h - 2^{-H} \leq f_{2^{-L}I, 2^{-L}I+a} \leq \sum_{h=L}^{H-1} \tilde{\rho}_h + \tilde{\sigma}_H + 2^{-H}.$$

If a varies, ρ_h varies among 2^h different functions and σ_H varies among 2^H different functions. By denoting them by $\rho_h^{(j)}$ ($1 \leq j \leq 2^h$), and by $\sigma_H^{(j)}$ ($1 \leq j \leq 2^H$), we introduce the following notation:

$$F(M, N, j, h)(x) := \begin{cases} |\sum_{k=M+1}^{M+N} \tilde{\rho}_h^{(j)}(\theta^k x)|, & \text{if } h < H, \\ |\sum_{k=M+1}^{M+N} \tilde{\sigma}_h^{(j)}(\theta^k x)|, & \text{if } h = H, \end{cases}$$

and $Y_{I', a', I, a}^{M, N}(x) := |\sum_{k=M+1}^{M+N} f_{2^{-L}I' + a', 2^{-L}I+a}(\theta^k x)|$. Then, we have

$$Y_{I, 0, I, a}^{M, N} \leq \sum_{h=L}^H F(M, N, j_h, h) + 2^{-H} N, \quad \text{for some } j_L, \dots, j_H.$$

By applying Lemma 4 of [5], for n satisfying $2^n \leq N < 2^{n+1}$, there exist $0 \leq m_l < 2^{n-l}$ ($1 \leq l \leq n$) such that

$$F(0, N, j, h) \leq F(0, 2^n, j, h) + \sum_{n/3 \leq l \leq n} F(2^n + m_l 2^l, 2^{l-1}, j, h) + N^{1/3}.$$

Thus we have

$$Y_{I, 0, I, a}^{0, N} \leq \sum_{h=L}^H \left(F(0, 2^n, j_h, h) + \sum_{n/3 \leq l \leq n} F(2^n + m_l 2^l, 2^{l-1}, j_h, h) \right) + 2\sqrt{N}.$$

Put $\phi_N := (2N \log \log N)^{1/2}$. Lemma 5 of [5] claims that, for any $\eta > 0$, there exist $n_0(\eta)$ and $C > 0$ such that $\lambda(E_{n_0(\eta)}) < \eta$ where

$$E_{n_0(\eta)} := \bigcup_{n \geq n_0(\eta)} \bigcup_{h \leq H} \bigcup_{j \leq 2^h} \left(G(n, j, h) \cup \bigcup_{n/3 \leq l \leq n} \bigcup_{m \leq 2^{n-l}} H(n, j, h, l, m) \right),$$

$$G(n, j, h) := \{x \mid F(0, 2^n, j, h) \geq C 2^{-h/8} \phi_{2^n}\},$$

$$H(n, j, h, l, m) := \{x \mid F(2^n + m 2^l, 2^{l-1}, j, h) \geq C 2^{-h/8} 2^{(l-n-3)/6} \phi_{2^n}\}.$$

By this, we see that $Y_{I,0,I,a}^{0,N}(x)$ is bounded from above by

$$\phi_{2^n} \sum_{h=L+1}^H 2^{-h/8} \left(1 + \sum_{n/3 \leq l \leq n} 2^{(l-n-3)/6} \right) + 2\sqrt{N} \leq 2^{5-L/8} \phi_N + 2\sqrt{N}$$

on $E_{n_0(\eta)}^c$. Since this estimate holds for all a , we have

$$\overline{\lim}_{N \rightarrow \infty} \sup_{a < 2^{-L}} Y_{I,0,I,a}^{0,N} / \phi_N \leq C 2^{5-L/8}$$

on $E_{n_0(\eta)}^c$, and hence on $\bigcup_k E_{n_0(1/k)}^c$, or for almost everywhere. Hence, we have

$$\overline{\lim}_{N \rightarrow \infty} \max_{I=0}^{2^L-1} \sup_{a < 2^{-L}} Y_{I,0,I,a}^{0,N} / \phi_N \leq C 2^{5-L/8} \xrightarrow{L \rightarrow \infty} 0, \quad \text{a.e.}$$

Applying (2.2) to $f_{2^{-L}I', 2^{-L}I}$ ($I' < I < 2^L$), we have

$$\overline{\lim}_{N \rightarrow \infty} \max_{I=0}^{2^L-1} Y_{0,0,I,0}^{0,N} / \phi_N = \max_{I=0}^{2^L-1} \sigma_{\theta,0,2^{-L}I} \xrightarrow{L \rightarrow \infty} \sup_{a < 1} \sigma_{\theta,0,a}, \quad \text{a.e.,}$$

$$\overline{\lim}_{N \rightarrow \infty} \max_{I' < I < 2^L} Y_{I',0,I,0}^{0,N} / \phi_N = \max_{I' < I < 2^L} \sigma_{\theta,2^{-L}I', 2^{-L}I} \xrightarrow{L \rightarrow \infty} \sup_{a' < a < 1} \sigma_{\theta,a',a}, \quad \text{a.e.,}$$

because of the continuity of $\sigma_{\theta,a',a}$ in a and a' , which will be proved in the next section. By applying $\sup_{a < 1} h(a) = \max_{I=0}^{2^L-1} \sup_{a < 2^{-L}} h(2^{-L}I + a)$, we have

$$\max_{I=0}^{2^L-1} Y_{0,0,I,0}^{0,N} \leq \sup_{a < 1} Y_{0,0,0,a}^{0,N} \leq \max_{I=0}^{2^L-1} Y_{0,0,I,0}^{0,N} + \max_{I=0}^{2^L-1} \sup_{a < 2^{-L}} Y_{I,0,I,a}^{0,N},$$

$$\max_{I' < I < 2^L} Y_{I',0,I,0}^{0,N} \leq \sup_{a' < a < 1} Y_{0,a',0,a}^{0,N} \leq \max_{I' < I < 2^L} Y_{I',0,I,0}^{0,N} + 2 \max_{I=0}^{2^L-1} \sup_{a < 2^{-L}} Y_{I,0,I,a}^{0,N}.$$

By gathering these results, we have

$$\overline{\lim}_{N \rightarrow \infty} \sup_{a' < a < 1} Y_{0,a',0,a}^{0,N} / \phi_N = \sup_{a' < a < 1} \sigma_{\theta,a',a}, \quad \overline{\lim}_{N \rightarrow \infty} \sup_{a < 1} Y_{0,0,0,a}^{0,N} / \phi_N = \sup_{a < 1} \sigma_{\theta,0,a}.$$

Hence we have completed the proof of Theorem except for the verification of

$$\sup_{a' < a < 1} \sigma_{\theta,a',a} = \sup_{a < 1} \sigma_{\theta,0,a}, \quad (3.1)$$

and the continuity of $\sigma_{\theta,a',a}$, which are proved in the next section.

4. Expression of $\sigma_{\theta,a',a}$

When $\theta^r \notin \mathbf{Q}$, we have $\sigma_{\theta,a',a}^2 = \int_0^1 f_{a',a}^2(x) dx = (a - a') - (a - a')^2$, and thereby $\sup_{a' < a < 1} \sigma_{\theta,a',a}^2 = \sup_{a < 1} \sigma_{\theta,0,a}^2 = 1/4$, and hence the assertion of the Theorem and (1) of the Corollary are proved.

We assume $\theta = \sqrt[r]{p/q}$ where $r = \min\{k \in \mathbf{N} \mid \theta^k \in \mathbf{Q}\}$ and $\gcd(p, q) = 1$, and prove

$$r_n^{a',a} := \int_0^1 f_{a',a}(p^n t) f_{a',a}(q^n t) dt = \tilde{V}(\langle p^n a' \rangle, \langle p^n a \rangle, \langle q^n a' \rangle, \langle q^n a \rangle) / p^n q^n,$$

where $\tilde{V}(x, y, \xi, \eta) := V(x, \xi) + V(y, \eta) - V(x, \eta) - V(y, \xi)$ and $V(x, \xi) := x \wedge \xi - x\xi$. This follows from the next lemma.

Lemma 1. *Let μ and ν be relatively prime positive integers. Then we have $r^{a',a} := \int_0^1 f_{a',a}(\mu t) f_{a',a}(\nu t) dt = \tilde{V}(\langle \mu a' \rangle, \langle \mu a \rangle, \langle \nu a' \rangle, \langle \nu a \rangle) / \mu\nu$.*

Proof: Denote $f := f_{a',a}$. Since the integrand has period 1,

$$r^{a',a} = \frac{1}{\mu\nu} \sum_{k=0}^{\nu-1} \sum_{j=0}^{\mu-1} \int_0^1 f(\mu(t + j/\mu + k/\nu)) f(\nu(t + j/\mu + k/\nu)) dt.$$

Thanks to $f(\mu(t + j/\mu + k/\nu)) = f(\mu(t + k/\nu))$ and $f(\nu(t + j/\mu + k/\nu)) = f(\nu(t + j/\mu))$, we have $r^{a',a} = \int_0^1 \Gamma_n(t) \Delta_n(t) dt / \mu\nu$, where $\Gamma_n(t) := \sum_{k=0}^{\nu-1} f(\mu(t + k/\nu))$ and $\Delta_n(t) := \sum_{j=0}^{\mu-1} f(\nu(t + j/\mu))$. Since μ and ν are relatively prime, we see that the transforms $k \mapsto \mu k$ on $\mathbf{Z}/\nu\mathbf{Z}$ and $j \mapsto \nu j$ on $\mathbf{Z}/\mu\mathbf{Z}$ are bijective. Hence we have $\Gamma_n(t) = \sum_{k=0}^{\nu-1} f(\mu t + k/\nu)$ and $\Delta_n(t) = \sum_{j=0}^{\mu-1} f(\nu t + j/\mu)$. Note that both of $1/\mu$ and $1/\nu$ are periods of $\Gamma_n \Delta_n$. Since μ and ν are relatively prime, we can find integers P and Q such that $1 = P\mu + Q\nu$. Thus we can conclude that $1/\mu\nu = P/\nu + Q/\mu$ is also a period, and hence

$$r^{a',a} = \int_0^{1/\mu\nu} \Gamma_n(t) \Delta_n(t) dt.$$

By $0 \leq t < 1/\mu\nu$, we have $0 \leq \mu t + k/\nu < 1/\nu$. Hence $\mathbf{1}_{[a',a)}(\mu t + k/\nu) = 1$ if and only if $[\nu a'] + \langle \nu a' \rangle \leq \mu\nu t + k < [\nu a] + \langle \nu a \rangle$. The last condition holds if and only if one of the following conditions is satisfied: (i) $[\nu a'] + 1 \leq k \leq [\nu a] - 1$, (ii) $k = [\nu a']$ and $\langle \nu a' \rangle / \mu\nu \leq t$, (iii) $k = [\nu a]$ and $t < \langle \nu a \rangle / \mu\nu$. Thus we have

$$\begin{aligned} \Gamma_n &= [\nu a] - [\nu a'] - 1 + \mathbf{1}_{[0, \langle \nu a \rangle / \mu\nu)} + \mathbf{1}_{[\langle \nu a' \rangle / \mu\nu, 1]} - \nu(a - a') \\ &= \mathbf{1}_{[0, \langle \nu a \rangle / \mu\nu)} - \mathbf{1}_{[0, \langle \nu a' \rangle / \mu\nu)} + \langle \nu a' \rangle - \langle \nu a \rangle, \\ \Delta_n &= [\mu a] - [\mu a'] - 1 + \mathbf{1}_{[0, \langle \mu a \rangle / \mu\nu)} + \mathbf{1}_{[\langle \mu a' \rangle / \mu\nu, 1]} - \mu(a - a') \\ &= \mathbf{1}_{[0, \langle \mu a \rangle / \mu\nu)} - \mathbf{1}_{[0, \langle \mu a' \rangle / \mu\nu)} + \langle \mu a' \rangle - \langle \mu a \rangle. \end{aligned}$$

By integrating $\Gamma_n \Delta_n$ over $[0, 1/\mu\nu)$, we have our expression. ■

By using the expression, let us prove the uniform continuity of $\sigma_{\theta, a', a}^2$ in (a', a) . Since $a \mapsto \langle p^n a \rangle$ is a continuous mapping from $\mathbf{R}$ to 1-dimensional torus $\mathbf{T} = \mathbf{R}/\mathbf{Z}$, and since $V(x, y)$ is a continuous function of $\mathbf{T} \times \mathbf{T}$, we see that $r_n^{a', a}$ is continuous in (a', a) and $|r_n^{a', a}| \leq 1/4p^n q^n$. Hence $\sigma_{\theta, a', a}^2$ is uniformly continuous in (a', a) .

Lemma 2. (3.1) is valid.

Proof: We prove $\widetilde{V}(\langle \mu a' \rangle, \langle \mu a \rangle, \langle \nu a' \rangle, \langle \nu a \rangle) \leq V(\langle \mu(a - a') \rangle, \langle \nu(a - a') \rangle)$ or

$$\begin{aligned} \widetilde{V}(\langle \mu a' \rangle, \langle \mu a \rangle, \langle \nu a' \rangle, \langle \nu a \rangle) &\leq \langle \mu a - \mu a' \rangle (1 - \langle \nu a - \nu a' \rangle), \\ \widetilde{V}(\langle \mu a' \rangle, \langle \mu a \rangle, \langle \nu a' \rangle, \langle \nu a \rangle) &\leq (1 - \langle \mu a - \mu a' \rangle) \langle \nu a - \nu a' \rangle, \end{aligned} \quad (4.1)$$

where μ and ν are positive integers and $a > a'$. Once it is proved, we have $r_n^{a', a} \leq r_n^{0, a-a'}$ and hence $\sigma_{\theta, a', a}^2 \leq \sigma_{\theta, 0, a-a'}^2$, which proves (3.1).

Denote $W(x, y, \xi, \eta) := (y - x)(\eta - \xi)$ and $\widetilde{W}(x, y, \xi, \eta) := x \wedge \xi + y \wedge \eta - x \wedge \eta - y \wedge \xi$. Clearly, $\widetilde{V} = \widetilde{W} - W$. We show one of the key inequalities:

$$0 \leq \widetilde{W}(x, y, \xi, \eta) \leq (x - y) \wedge (\xi - \eta), \quad (\xi \geq \eta, x \geq y). \quad (4.2)$$

It is verified by $0 \leq y \wedge \xi - y \wedge \eta \leq x \wedge \xi - x \wedge \eta \leq \xi - \eta$, and by symmetry. By exchanging x and y in the last inequality, we have the other key inequality:

$$\widetilde{W}(x, y, \xi, \eta) \leq 0, \quad (\xi \geq \eta, x \leq y). \quad (4.3)$$

Recall that $\langle x \rangle \geq \langle y \rangle$ implies $\langle x - y \rangle = \langle x \rangle - \langle y \rangle$, and $\langle x \rangle < \langle y \rangle$ implies $\langle x - y \rangle = \langle x \rangle - \langle y \rangle + 1$.

We denote $W(\langle \mu a' \rangle, \langle \mu a \rangle, \langle \nu a' \rangle, \langle \nu a \rangle)$ etc. simply by W and $\widetilde{W}$.

If $\langle \mu a \rangle \geq \langle \mu a' \rangle$ and $\langle \nu a \rangle \geq \langle \nu a' \rangle$, by (4.2) we have $\widetilde{W} \leq \langle \mu a \rangle - \langle \mu a' \rangle = \langle \mu a - \mu a' \rangle$ and $\widetilde{W} \leq \langle \nu a \rangle - \langle \nu a' \rangle = \langle \nu a - \nu a' \rangle$. By adding $-W = -(\langle \mu a - \mu a' \rangle)(\langle \nu a - \nu a' \rangle)$ to these, we have (4.1).

If $\langle \mu a \rangle < \langle \mu a' \rangle$ and $\langle \nu a \rangle \geq \langle \nu a' \rangle$, we have $-W = -(\langle \mu a - \mu a' \rangle - 1)\langle \nu a - \nu a' \rangle$. We have $\widetilde{W} \leq 0$ by (4.3), and $\widetilde{W} \leq \langle \mu a \rangle + \langle \nu a' \rangle - \langle \mu a \rangle \wedge \langle \nu a' \rangle - \langle \mu a' \rangle \wedge \langle \nu a \rangle$ by definition. The right hand side equals to one of the following, which are all bounded by $\zeta := \langle \mu a - \mu a' \rangle - \langle \nu a - \nu a' \rangle$:

$$\begin{aligned} \langle \mu a \rangle - \langle \mu a' \rangle &= \langle \mu a - \mu a' \rangle - 1 \leq \langle \mu a - \mu a' \rangle - \langle \nu a - \nu a' \rangle = \zeta, \\ \langle \mu a \rangle - \langle \nu a \rangle &= \langle \mu a - \mu a' \rangle - \langle \nu a - \nu a' \rangle - 1 + \langle \mu a' \rangle - \langle \nu a' \rangle \leq \zeta, \\ \langle \nu a' \rangle - \langle \mu a' \rangle &= \langle \mu a - \mu a' \rangle - \langle \nu a - \nu a' \rangle - 1 - \langle \mu a \rangle + \langle \nu a \rangle \leq \zeta, \\ \langle \nu a' \rangle - \langle \nu a \rangle &= -\langle \nu a - \nu a' \rangle \leq \langle \mu a - \mu a' \rangle - \langle \nu a - \nu a' \rangle = \zeta. \end{aligned}$$

These estimates imply (4.1).

If $\langle \mu a \rangle \geq \langle \mu a' \rangle$ and $\langle \nu a \rangle < \langle \nu a' \rangle$, then we have (4.1) by $\tilde{V}(x, y, \xi, \eta) = \tilde{V}(\xi, \eta, x, y)$ and by the fact just we have proved.

If $\langle \mu a \rangle < \langle \mu a' \rangle$ and $\langle \nu a \rangle < \langle \nu a' \rangle$, by $\tilde{V}(x, y, \xi, \eta) = \tilde{V}(y, x, \eta, \xi)$ and the fact that we have proved, we can prove that $\tilde{V}$ is bounded by both of $\langle \mu a' - \mu a \rangle (1 - \langle \nu a' - \nu a \rangle)$ and $(1 - \langle \mu a' - \mu a \rangle) \langle \nu a' - \nu a \rangle$. By noting $\langle -x \rangle = 1 - \langle x \rangle$, we have (4.1). ■

Lemma 3. *The symmetry $\sigma_{\theta,0,a} = \sigma_{\theta,0,1-a}$ holds.*

Proof: Thanks to $\langle \mu(1-a) \rangle = \langle -\mu a \rangle = 1 - \langle \mu a \rangle$ and $V(1-x, 1-y) = V(x, y)$, we have $V(\langle \mu(1-a) \rangle, \langle \nu(1-a) \rangle) = V(\langle \mu a \rangle, \langle \nu a \rangle)$. By Lemma 1, we complete the proof. ■

5. Calculation of Σ_θ .

Lemma 1 yields $\sigma_{\theta,0,a}^2 = V(a, a) + 2 \sum_{n=1}^{\infty} V(\langle p^n a \rangle, \langle q^n a \rangle) / p^n q^n$. By noting $0 \leq V(x, \xi) \leq V(\xi, \xi) = \xi - \xi^2 \leq 1/4 = V(1/2, 1/2)$ ($x, \xi \in [0, 1)$), we have

$$\sigma_{\theta,0,a}^2 \leq \frac{1}{4} + 2 \sum_{n=1}^{\infty} \frac{1}{4p^n q^n} = \frac{pq+1}{4(pq-1)} \quad \text{and} \quad \sigma_{\theta,0,1/2}^2 \geq V(1/2, 1/2) = \frac{1}{4}.$$

These prove (2) of Corollary.

When p and q are odd, thanks to $\langle p^n/2 \rangle = \langle q^n/2 \rangle = 1/2$, we have

$$\sigma_{\theta,0,1/2}^2 = V(1/2, 1/2) + 2 \sum_{n=1}^{\infty} \frac{V(1/2, 1/2)}{p^n q^n} = \frac{pq+1}{4(pq-1)},$$

which is nothing but the assertion of (3). (4) is clear from (3).

Here after, let us assume $q = 1$, and that $p \geq 4$ is even. Since $\sigma_{\theta,0,a}$ does not depend on r , it is sufficient to study $\sigma_{p,0,a}$. We prove

$$\sup_{a < 1} \sigma_{p,0,a}^2 = \sigma_{p,0,a_p}^2 = \frac{(p+1)p(p-1)(p-2)}{4(p-1)^4}, \quad \text{where} \quad a_p = \frac{p/2-1}{p-1}.$$

By Lemma 3, it is enough to prove $\sigma_{p,0,a}^2 \leq \sigma_{p,0,a_p}^2$ for $0 \leq a \leq 1/2$.

Because of $V(\langle p^n a \rangle, a) \leq V(a, a)$, we have $\sigma_{p,0,a}^2 \leq V(a, a)(p+1)/(p-1)$. Since $\langle p^n a_p \rangle = a_p$, we see that $\sigma_{p,0,a_p}^2 = V(a_p, a_p)(p+1)/(p-1)$. Thus $a \leq a_p$ implies $V(a, a) \leq V(a_p, a_p)$ and $\sigma_{p,0,a}^2 \leq \sigma_{p,0,a_p}^2$.

Let $a_p \leq a \leq 1/2$. The estimate $(p/2-1)/p < a_p$ implies $\langle pa \rangle = pa - p/2 + 1 \geq a$, and hence $V(\langle pa \rangle, a) = a - a(pa - p/2 + 1)$ and thereby

$$\sigma_{p,0,a}^2 \leq V(a, a) + \frac{2}{p}(a - a(pa - p/2 + 1)) + 2 \sum_{n=2}^{\infty} \frac{V(a, a)}{p^n} =: h_1(a).$$

Note that the equality holds if $a = a_p$, i.e., $\sigma_{p,0,a_p}^2 = h_1(a_p)$.

We here assume $p \geq 6$ and prove that $h_1(a)$ decreases for $a \geq a_p$. By these we can conclude that $\sigma_{p,0,a}^2 \leq h_1(a) \leq h_1(a_p) = \sigma_{p,0,a_p}^2$ for $1/2 \geq a \geq a_p$, and complete the proof of the case $p \geq 6$.

Because $h_1(a) = a(2(p^2 - p + 1) - (3p^2 - 3p + 2)a)/p(p - 1)$ decreases for $a \geq (p^2 - p + 1)/(3p^2 - 3p + 2)$, it is enough to prove that

$$a_p - \frac{p^2 - p + 1}{3p^2 - 3p + 2} = \frac{p^3 - 5p^2 + 4p - 2}{2(p - 1)(3p^2 - 3p + 2)} =: D_1$$

is positive. Since $\frac{d}{dp}(p^3 - 5p^2 + 4p - 2) = 3p^2 - 10p + 4$ increases for $p \geq 5/3$ and $p^3 - 5p^2 + 4p - 2|_{p=6} > 0$, we can conclude that $p^3 - 5p^2 + 4p - 2 > 0$ for $p \geq 6$, and hence $D_1 > 0$ for $p \geq 6$.

Next, let us examine the case $p = 4$. By the above argument we have $\sigma_{4,0,a}^2 \leq \sigma_{4,0,a_4}^2 = h_1(a_4)$ for $a \leq a_4 = 1/3$. Because $h_1(a) = a(13 - 19a)/6$ decreases for $a > 3/8$ we see that $\sigma_{4,0,a}^2 \leq h_1(a) \leq h_1(3/8) = 47/128 < 10/27 = h_1(a_4) = \sigma_{4,0,a_4}^2$ for $8/3 \leq a \leq 1/2$. Therefore it is enough to consider the case $1/3 \leq a < 3/8$. In this case we have $\langle 4^2 a \rangle = 16a - 5 \geq a$, and hence $V(\langle 4^2 a \rangle, a) = a - a(16a - 5)$ and

$$\sigma_{4,0,a}^2 \leq V(a, a) + \frac{2}{4}(a - a(4a - 1)) + \frac{2}{16}(a - a(16a - 5)) + 2 \sum_{n=3}^{\infty} \frac{V(a, a)}{4^n} =: h_2(a).$$

Note that the equality holds for $a = a_4$, i.e., $\sigma_{4,0,a_4}^2 = h_2(a_4)$. Since $h_2(a) = a(67 - 121a)/24$ decreases for $a \geq a_4$, we have $\sigma_{4,0,a}^2 \leq h_2(a) \leq h_2(a_4) = \sigma_{4,0,a_4}^2$ for $1/3 \leq a < 3/8$. This complete the proof of the case $p = 4$. Therefore (5) is proved.

Now let us prove (6). We prove $\sigma_{2,0,a}^2 \leq \sigma_{2,0,1/3}^2$. To begin with, we prepare a lemma.

Lemma 4. For $0 \leq a \leq 1/2$ and $0 \leq a' < 1$, it holds that

$$V^*(a', a) := V(a', a) + \frac{1}{2}V(\langle 2a' \rangle, a) \leq V(a, a) + \frac{1}{2}V(2a, a).$$

When $a' = a$, the equality holds.

Proof: We have $V^*(a', a) = a'(1 - a)$ if $0 \leq a' \leq a/2$, $V^*(a', a) = a'(1 - 2a) + a/2$ if $a/2 \leq a' \leq a$, and $V^*(a', a) = a'(1 - 2a) + 3a/2 - 1/2$ if $1/2 \leq a' \leq (a + 1)/2$, in all cases V^* increases in a' . On the other hand, $V^*(a', a) = -2aa' + 3a/2$ if $a \leq a' \leq 1/2$, and $V^*(a', a) = -2aa' + 2a$ if $(a + 1)/2 \leq a' \leq 1/2$, in both cases V^* decreases in a' . Hence, for given a , $V^*(\cdot, a)$ takes maximal values at $a' = a$ and $a' = (a + 1)/2$. Because of

$$V^*(a, a) = V(a, a) + \frac{1}{2}V(2a, a) \geq V\left(\frac{a + 1}{2}, a\right) + \frac{1}{2}V(a, a) = V^*\left(\frac{a + 1}{2}, a\right),$$

we have $\max_{a'} V^*(a', a) = V^*(a, a)$. ■

By this lemma, we have $V(\langle 2^{2n}a \rangle, a) + V(\langle 2^{2n+1}a \rangle, a)/2 \leq V^*(a, a)$ for $0 \leq a \leq 1/2$. The equality holds for $a = 1/3$ since $\langle 2^{2n}/3 \rangle = 1/3$. Hence we have

$$\sigma_{2,0,a}^2 \leq V(a, a) + V(2a, a) + 2 \sum_{n=1}^{\infty} \frac{V^*(a, a)}{4^n} =: h_3(a),$$

where the equality holds for $a = 1/3$. Since $h_3(a) = a(9 - 13a)/3$ decreases for $a \leq 1/3$, we see $\sigma_{2,0,a}^2 \leq \sigma_{2,0,1/3}^2$ for $a \leq 1/3$. Since h_3 decreases for $a \geq 3/8$ and $h_3(3/8) = 33/64 < 14/27 = h_3(1/3) = \sigma_{2,0,1/3}^2$, we have $\sigma_{2,0,a}^2 \leq \sigma_{2,0,1/3}^2$ for $3/8 \leq a \leq 1/2$. When $1/3 < a < 3/8$, we have $\sigma_{2,0,a}^2 \leq h_4(a)$, where

$$h_4(a) := V(a, a) + V(2a, a) + \frac{V(4a - 1, a)}{2} + \frac{V(8a - 2, a)}{4} + 2 \sum_{n=2}^{\infty} \frac{V^*(a, a)}{4^n}.$$

The equality holds for $a = 1/3$. Since $h_4(a) = 2a(6 - 11a)/3$ decreases for $a \geq 1/3$, we see that $\sigma_{2,0,a}^2 \leq h_4(1/3) = \sigma_{2,0,1/3}^2$ for $1/3 < a < 3/8$. This completes the proof of (6).

Lastly, we present another method to calculate the supremum of $\sigma_{p/q,0,a}$. Unfortunately, it seems difficult to apply this method generally, and we consider only the special case $p/q = 5/2$, which is (7).

Because of $\langle 5^n/3 \rangle = \langle 2^n/3 \rangle = 1/3, 2/3$, we have $V(\langle 2^n/3 \rangle, \langle 5^n/3 \rangle) = 2/9$, and hence we have $\sigma_{5/2,0,1/3}^2 = (2/9)(1 + 2 \sum_{n=1}^{\infty} 1/10^n) = 22/81$. We prove that this is the maximum of $\sigma_{5/2,0,a}^2$.

Put $I_n = ((1 - 1/5^n)/3, (1 + 2/5^n)/3)$ and $J_n = ((1 - 1/2^n)/3, (1 + 2/2^n)/3)$ if n is even, and put $I_n = ((1 - 2/5^n)/3, (1 + 1/5^n)/3)$ and $J_n = ((1 - 2/2^n)/3, (1 + 1/2^n)/3)$ otherwise. Clearly we have $1/3 \in I_n \subset J_n$, $I_{n+1} \subset I_n$ and $\bigcap I_n = \{1/3\}$. $\langle 5^n a \rangle$ is linear on I_n , and $\langle 2^n a \rangle$ is linear on J_n and hence on I_n .

If $a \in I_n$ and $a < 1/3$, we have $V(\langle 2^n a \rangle, \langle 5^n a \rangle) = \langle 5^n a \rangle(1 - \langle 2^n a \rangle)/10^n$ and hence $V(\langle 2^n a \rangle, \langle 5^n a \rangle)' = 5^n - 5^n \langle 2^n a \rangle - 2^n \langle 5^n a \rangle \geq 5^n - 5^n \langle 2^n/3 \rangle - 2^n \langle 5^n/3 \rangle \geq 5^n - 2(5^n + 2^n)/3 > 0$.

If $a \in I_n$ and $a > 1/3$, we have $V(\langle 2^n a \rangle, \langle 5^n a \rangle) = \langle 2^n a \rangle(1 - \langle 5^n a \rangle)/10^n$ and hence $V(\langle 2^n a \rangle, \langle 5^n a \rangle)' = 2^n - 5^n \langle 2^n a \rangle - 2^n \langle 5^n a \rangle \leq 2^n - 5^n \langle 2^n/3 \rangle - 2^n \langle 5^n/3 \rangle \leq 2^n - (5^n + 2^n)/3 < 0$.

Put $\tau_n(a) = V(a, a) + 2 \sum_{k=1}^n V(\langle 2^k a \rangle, \langle 5^k a \rangle)/10^k$. Clearly, $\tau_{\infty}(a) = \sigma_{5/2,0,a}^2$. When $a \in I_n$ we have $\tau_n' \geq V(a, a)' = 1 - 2a \geq 1/3$ if $a < 1/3$, and $\tau_n' \leq V(a, a)' + V(\langle 2a \rangle, \langle 5a \rangle)'/5 \leq -1/5$ if $a > 1/3$. Hence we have $\tau_n(1/3) - \tau_n(a) \geq |a - 1/3|/5 \geq 1/(45 \cdot 5^n)$ for $a \in I_n \setminus I_{n+1}$. By summing it with $-\tau_{\infty}(a) + \tau_n(a) \geq -2 \sum_{k=n+1}^{\infty} 1/(4 \cdot 10^k) = -1/(18 \cdot 10^n)$ and $\tau_{\infty}(1/3) -$

$\tau_n(1/3) = 2 \sum_{k=n+1}^{\infty} 2/(9 \cdot 10^k) = 4/(81 \cdot 10^n)$, we have

$$\tau_{\infty}(1/3) - \tau_{\infty}(a) \geq \frac{1}{45 \cdot 5^n} - \frac{1}{18 \cdot 10^n} + \frac{4}{81 \cdot 10^n} = \frac{1}{45 \cdot 5^n} - \frac{1}{162 \cdot 10^n} > 0,$$

or $\tau_{\infty}(a) < \tau_{\infty}(1/3)$ on $I_n \setminus I_{n+1}$ and hence on $\bigcup I_n \setminus I_{n+1} = I_1 \setminus \{1/3\}$.

Since $\tau_1(a)$ is piecewise quadratic function, elementary calculus shows that $\tau_1(a) \leq 19/75$ on I_1^c . Thus we can verify $\tau_{\infty}(a) < \tau_{\infty}(1/3)$ on I_1^c as below:

$$\tau_{\infty}(1/3) - \tau_{\infty}(a) = \tau_n(1/3) - \tau_1(a) - \tau_{\infty}(a) + \tau_1(a) \geq \frac{22}{81} - \frac{19}{75} - \frac{1}{180} = \frac{1}{75}.$$

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